4. Several adverse scenarios were considered to further assess the sustainability of government debt. These standardized scenarios, which are applied to the baseline, include low growth, a deterioration in the primary balance, realization of contingent liabilities, and high borrowing costs. The results indicate that public debt is highly vulnerable to adverse macroeconomic conditions:

- *Real growth shock.* Real GDP growth rate is assumed to decline by one-standard deviation below the baseline during 2019–20 and the interest rate increase by 25 basis points for one percent of GDP deterioration of the primary balance compared to the baseline. Under this scenario, the public debt ratio would increase to 82 percent of GDP by 2020, and to 90 percent of GDP by 2024.

- *Primary balance shock.* A deterioration of 1.5 percentage points of GDP in the primary balance in both 2019–20 is expected to increase the public debt by 3 percentage points relative to the baseline, reaching 81 percent of GDP by 2024.

- *Contingent liability shock.* Additional costs related to banking system repair (10 percent of GDP) accompanied by lower growth in 2019–20 and higher borrowing cost (by additional 300 basis points) will increase public debt rapidly in an unsustainable manner to about 100 percent of GDP by 2024. Gross financing needs increase to about 9 percent of GDP.

- *Real interest rate shock.* Under this scenario, which envisages an interest rate increase of 300 basis points, public debt is projected to increase to 80 percent of GDP by 2024.

- *Combined macro-fiscal shock*. Under this scenario, which is a combination of the above shocks (excluding the contingent liability shock), the public debt would reach 84 percent of GDP by 2020, and then gradually increase to 93 percent of GDP by 2024.

5. San Marino's gross external debt is set to decline on the back of continued, albeit more moderate, capital outflows. San Marino's gross external debt stood at 89 percent of GDP in 2017, where deposit liabilities of deposit-taking corporations accounted for the largest component. With continued capital outflows, including from the banking system, gross external debt is projected to have declined to 78 percent of GDP in 2018 and remain on a downward trajectory over the medium term, reaching a level of 53 percent of GDP by 2024. All else equal, lower growth, a deterioration in the current account, and a slower price growth relative to that in key trading partners (real exchange rate depreciation) would result in a higher trajectory of external debt relative to the baseline.

6. The analysis is subject to a number of caveats reflecting limited data availability. First, due to lack of data, gross external debt does not include FDI debt instrument liabilities, and as a result, gross financing needs may be underestimated. In addition, limited availability of external sector historical data required making assumptions about averages and standard deviations of key variables, such as non-interest current account balance, export and import growth, and non-debt creating capital flows, that are used for the stress tests.

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Figure A5.1. San Marino: Public Sector Debt Sustainability Analysis (DSA)—Baseline Scenario

(In percent of GDP unless otherwise indicated)

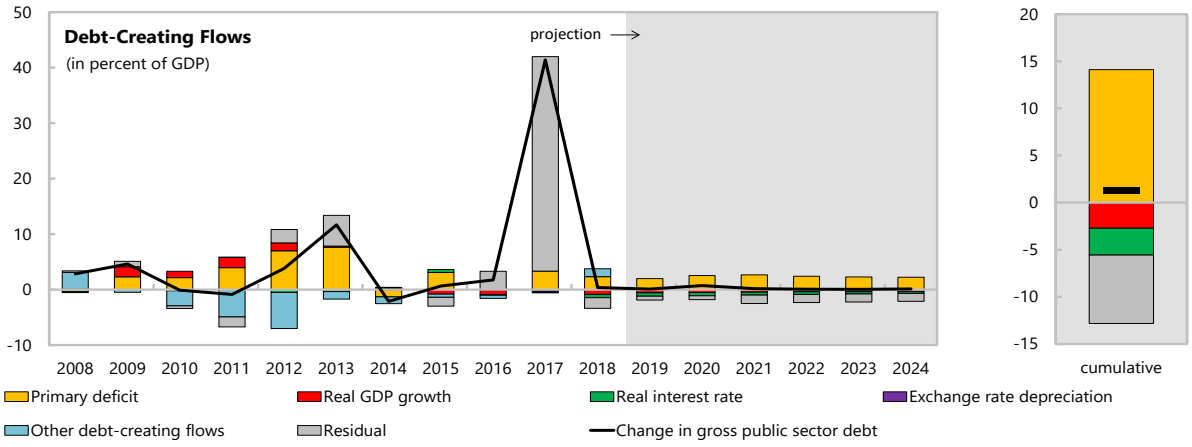
Debt, Economic and Market Indicators [1/]

	Actual			Projections					
	2011-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024
Nominal gross public debt	29.9	76.6	77.0	77.1	77.9	78.0	78.1	78.2	78.3
Public gross financing needs	4.2	4.5	3.9	3.9	4.9	5.6	5.8	6.1	6.5
Real GDP growth (in percent)	-2.0	0.6	1.1	0.8	0.7	0.6	0.5	0.5	0.5
Inflation (GDP deflator, in percent)	1.0	1.4	1.3	1.3	1.4	1.5	1.5	1.5	1.5
Nominal GDP growth (in percent)	-1.0	2.0	2.5	2.1	2.1	2.1	2.0	2.1	2.1
Effective interest rate (in percent) [4/]	0.6	0.8	0.5	0.6	0.7	0.8	0.9	1.0	1.2

As of December 03, 2018		
Sovereign Spreads		
EMBIG (bp) 3/		n.a.
5Y CDS (bp)		n.a.
Ratings	Foreign	Local
Moody's	n.a.	n.a.
S&Ps	n.a.	n.a.
Fitch	BBB-	BBB-

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	2.5	41.4	0.4	0.1	0.7	0.2	0.1	0.1	0.1	1.3	
Identified debt-creating flows	1.2	2.7	2.3	0.8	1.4	1.7	1.6	1.5	1.6	8.5	
Primary deficit	3.4	3.3	2.3	2.0	2.5	2.7	2.4	2.3	2.2	14.1	-0.7
Primary (noninterest) revenue and grants	23.0	22.1	22.9	20.9	20.9	20.6	20.7	20.7	20.6	124.5	
Primary (noninterest) expenditure	26.4	25.4	25.3	22.9	23.5	23.3	23.1	23.0	22.9	138.6	
Automatic debt dynamics [5/]	0.3	-0.4	-1.5	-1.2	-1.1	-1.0	-0.9	-0.8	-0.7	-5.6	
Interest rate/growth differential [6/]	0.3	-0.4	-1.5	-1.2	-1.1	-1.0	-0.9	-0.8	-0.7	-5.6	
Of which: real interest rate	0.0	-0.2	-0.6	-0.6	-0.6	-0.5	-0.5	-0.4	-0.3	-2.8	
Of which: real GDP growth	0.4	-0.2	-0.8	-0.6	-0.5	-0.5	-0.4	-0.4	-0.4	-2.7	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	-2.5	-0.2	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Drawdown of deposits and other operat	-2.5	-0.2	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	1.3	38.6	-1.9	-0.7	-0.7	-1.5	-1.5	-1.5	-1.4	-7.3	



Source: IMF staff.

1/ Public sector is defined as central government.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

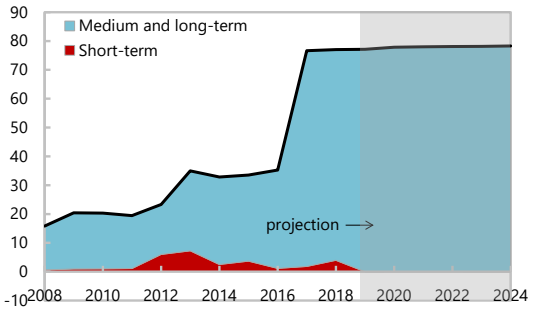
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Figure A5.2. San Marino: Public DSA—Composition of Public Debt and Alternate Scenarios

Composition of Public Debt

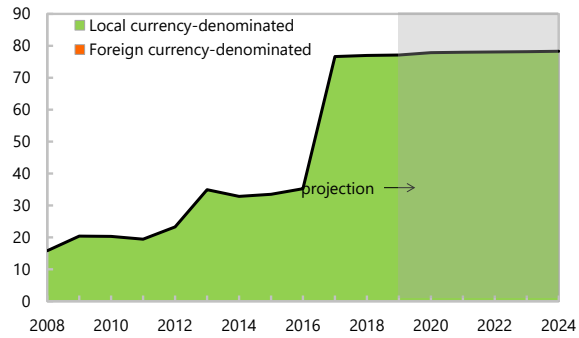
By Maturity

(in percent of GDP)



By Currency

(in percent of GDP)

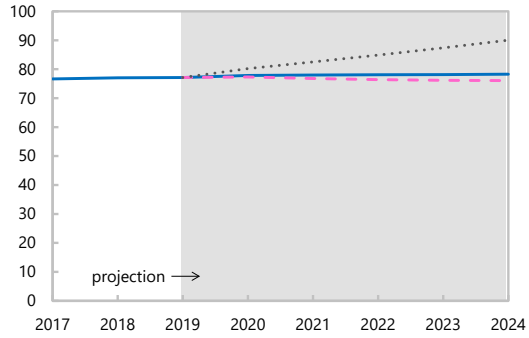


Alternative Scenarios



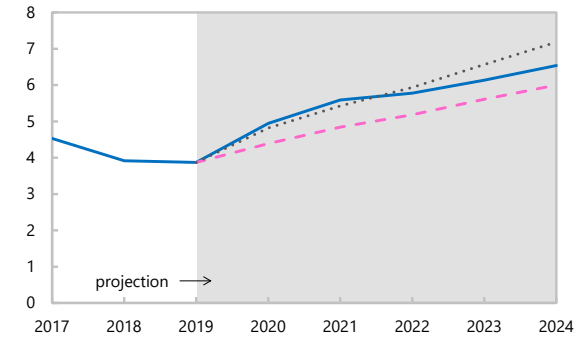
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	0.8	0.7	0.6	0.5	0.5	0.5
Inflation	1.3	1.4	1.5	1.5	1.5	1.5
Primary Balance	-2.0	-2.5	-2.7	-2.4	-2.3	-2.2
Effective interest rate	0.6	0.7	0.8	0.9	1.0	1.2
Constant Primary Balance Scenario						
Real GDP growth	0.8	0.7	0.6	0.5	0.5	0.5
Inflation	1.3	1.4	1.5	1.5	1.5	1.5
Primary Balance	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0
Effective interest rate	0.6	0.7	0.8	0.9	1.0	1.1

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	0.8	-2.6	-2.6	-2.6	-2.6	-2.6
Inflation	1.3	1.4	1.5	1.5	1.5	1.5
Primary Balance	-2.0	-2.3	-2.3	-2.3	-2.3	-2.3
Effective interest rate	0.6	0.7	0.8	0.9	1.0	1.2

Source: IMF staff.

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